

Energy-Aware Scheduling for Edge Computing Systems

A Multi-Objective Perspective on Latency, Energy, and Resource Utilization

Abstract

Edge computing moves computation closer to data sources, reducing communication latency but creating difficult resource-management problems. This paper examines energy-aware task scheduling as a multi-objective optimization problem involving latency, energy consumption, processor utilization, and deadline satisfaction. A scheduling framework is proposed in which tasks are dynamically assigned to heterogeneous edge nodes according to estimated execution cost and communication delay. The paper compares greedy priority scheduling with a weighted optimization formulation and discusses the trade-offs created by different objective weights. The analysis shows that minimizing latency alone can lead to inefficient energy use, while aggressive energy minimization may increase deadline violations.

Keywords: edge computing; task scheduling; energy efficiency; optimization; Internet of Things

1. Introduction

Edge computing supports applications such as industrial monitoring, autonomous systems, and interactive services where sending every task to a distant cloud can introduce unacceptable latency. However, edge nodes often have limited computational and energy resources. Efficient task scheduling is consequently a central research problem.

This paper investigates scheduling as a multi-objective problem. The objective is not simply to finish tasks quickly, but to find assignments that jointly control response time and energy consumption while satisfying service constraints.

2. Problem Formulation

Let a set of tasks be represented by T and a set of edge nodes by E . Each task has an estimated computational workload, input size, and deadline. Each edge node has a processing capacity and an energy-consumption profile. A scheduling decision determines the node to which each task is assigned.

A simplified objective can be expressed as $J = \alpha L + \beta E + \gamma V$, where L denotes normalized latency, E denotes normalized energy consumption, and V represents deadline violations. The weights α , β , and γ encode application priorities.

3. Scheduling Strategy

The proposed framework first estimates communication and execution costs for candidate nodes. In a greedy baseline, tasks are ordered according to urgency and assigned to the currently feasible node with the lowest estimated completion time. A multi-objective strategy instead evaluates a weighted score incorporating latency and energy.

Because edge environments change over time, scheduling decisions should be recalculated when node load, network quality, or task urgency changes. This dynamic aspect distinguishes the problem from static scheduling in which all workloads are known in advance.

4. Evaluation and Discussion

A representative simulation can vary the number of edge nodes, task arrival rates, processor heterogeneity, and network conditions. Results should report average latency, energy per completed task, deadline-violation rate, and node utilization. Under light load, both strategies may perform similarly. Under heavy load, however, a latency-only strategy can concentrate work on fast nodes and create hotspots.

The multi-objective formulation can distribute tasks more evenly and reduce energy consumption, although its benefits depend strongly on the selected weights. This demonstrates why scheduling policies should be tuned to application requirements rather than evaluated using one universal objective.

5. Conclusion

Energy-aware edge scheduling requires explicit consideration of competing objectives. A multi-objective formulation provides a principled way to trade latency against energy and service quality. Future work should incorporate reinforcement learning, renewable-energy availability, task migration costs, and real hardware measurements.

References

- Bengio, Y., Goodfellow, I., & Courville, A. (2016). *Deep Learning*. MIT Press.
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- Vaswani, A., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*.